Name: Irisha Deshpande
Class: T1

Roll No: 22
Subject: Agile Software Development and DevOps Lab

Experiment 3

Aim: To deploy multi service applications using Docker Compose

This experiment maps to LO4.
LO4:Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

Theory:

Docker Compose is a tool used to define and manage multiple Docker containers using a single YAML configuration file. It allows developers to create multi-container applications where different services can work together efficiently. In this experiment, a multi-service application is created using MySQL as the database service and phpMyAdmin as a web-based database management interface. The docker-compose.yml file contains the configuration for images, ports, environment variables, volumes, and service dependencies. Docker Compose automatically creates a network that enables communication between the MySQL and phpMyAdmin containers. MySQL is responsible for storing and managing structured data, while phpMyAdmin provides a graphical interface to perform database operations. Environment variables are used to configure database usernames, passwords, and initial database settings. Docker volumes are used to provide persistent storage so that MySQL data remains available even after containers are stopped or removed. Docker Compose simplifies the process of deploying, running, and managing multiple interconnected services. This approach is widely used for developing, testing, and deploying modern containerized applications.

Services Created Using Docker Compose

1. Jira Service
    - Jira is the main application service used for project management and issue tracking.
    - It provides features for creating, assigning, and monitoring tasks and software development activities.
    - It runs inside a Docker container and connects to the MySQL database for storing application data.
2. MySQL Service
    - MySQL is the database service used to store Jira data such as users, projects, issues, and configurations.
    - It provides a relational database environment required by Jira.
    - Docker volumes are used to maintain persistent storage of database information.
3. phpMyAdmin Service

- phpMyAdmin is a web-based database management service used to interact with MySQL through a graphical interface.
- It allows users to view, create, modify, and manage databases easily.
- It connects to the MySQL service through the Docker Compose network.

Code:

```
version: "3.9"


services:
  mysql:
    image: mysql:8.0
    container_name: mysql-db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: studentdb
      MYSQL_USER: student
      MYSQL_PASSWORD: student123
    ports:
      - "3306:3306"
    volumes:
      - mysql_data:/var/lib/mysql


  phpmyadmin:
    image: phpmyadmin/phpmyadmin
    container_name: phpmyadmin
```

```
    restart: always
    ports:
      - "8080:80"
    environment:
      PMA_HOST: mysql
      MYSQL_ROOT_PASSWORD: root123
    depends_on:
      - mysql

volumes:
  mysql_data:
```

Commands:
`docker compose up - Start all services
`docker compose up -d - Start in detached mode
`docker compose ps - Show running services
`docker compose logs - View logs
`docker compose stop- Stop services
`docker compose start Restart services
`docker compose restart- Restart containers
`docker compose down - Stop and remove containers
`docker compose down -v - Remove containers and volumes
`docker compose images - Show images
`docker compose config - Validate Compose file

Output

```
PS C:\Users\15L\Desktop\exp3> docker compose up -d
[+] Running 22/1
 ✓ phpmyadmin 21 layers [⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿⣿]      0B/0B      Pulled
[+] Running 0/3
 - Network exp3_default      Created
 - Volume "exp3_mysql_data"  Created
 - Container mysql-db        Starting
 - Container phpmyadmin      Created
Error response from daemon: Ports are not available: exposing port TCP 0.0.0.0:3
ch socket address (protocol/network address/port) is normally permitted.
PS C:\Users\15L\Desktop\exp3> docker ps
```

```
PS C:\Users\15L\Desktop\exp3> docker ps
CONTAINER ID   IMAGE           COMMAND                  CREATED         STATUS         PORTS     NAMES
23cb88a8c65b   alpine:latest   "sh -c 'while true; …"   5 minutes ago   Up 4 minutes             docker_expt_2-app-1
PS C:\Users\15L\Desktop\exp3> docker logs -f mysql-db
PS C:\Users\15L\Desktop\exp3> docker logs -f phpmyadmin
```

Database setup

Language

Learn more about connecting Jira to a database.

Database Type: MySQL 8.0

Jira requires that you download and install the MySQL 8.0 driver. You will have to restart Jira after installing the driver. Please consult our documentation for more information.

Hostname: mysql
Hostname or IP address of the database server.

Port: 3306
TCP Port Number for the database server.

Database: studentdb
The name of the database to connect to.

Username: root
The username used to access the database.

Password: •••••••
The password used to access the database.

Next | Test Connection

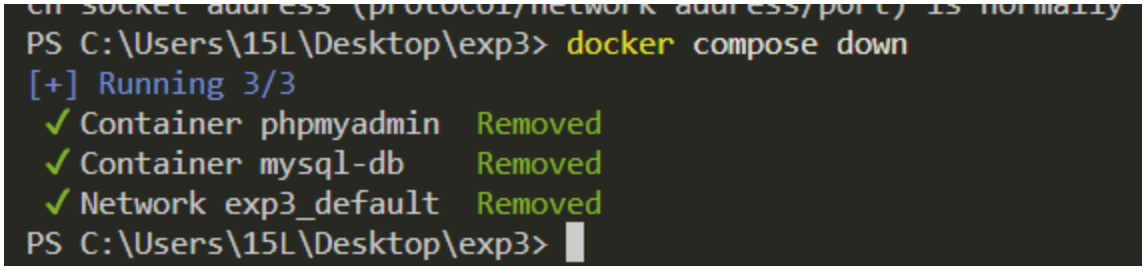
```
PS C:\Users\15L\Desktop\exp3> docker compose images
CONTAINER           REPOSITORY              TAG                 IMAGE ID            SIZE
mysql-db            mysql                   8.0                 6cd09145362d        799MB
phpmyadmin          phpmyadmin/phpmyadmin   latest              e66b1f5a8c58        742MB
```

```
PS C:\Users\15L\Desktop\exp3> docker compose config
name: exp3
services:
  mysql:
    container_name: mysql-db
    environment:
      MYSQL_DATABASE: studentdb
      MYSQL_PASSWORD: student123
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_USER: student
    image: mysql:8.0
    networks:
      default: null
    ports:
      - mode: ingress
        target: 3306
        published: "3306"
        protocol: tcp
    restart: always
    volumes:
      - type: volume
        source: mysql_data
```

```
PS C:\Users\15L\Desktop\exp3> docker compose restart
[+] Restarting 2/2
 ✓ Container phpmyadmin  Started
 ✓ Container mysql-db    Started
PS C:\Users\15L\Desktop\exp3>
```

```
PS C:\Users\15L\Desktop\exp3> docker compose stop
[+] Stopping 2/2
 ✓ Container phpmyadmin  Stopped
 ✓ Container mysql-db    Stopped
```

```
PS C:\Users\15L\Desktop\exp3> docker compose start
[+] Running 0/0
 - Container mysql-db  Starting
Error response from daemon: Ports are not available: exposing port TCP 0.0.0.0:3306 -> 0.0.0.0:0: listen tcp 0.0.0.0:3306:
ch socket address (protocol/network address/port) is normally permitted.
PS C:\Users\15L\Desktop\exp3>
```



Conclusion:

In this experiment, the learner understood how to deploy and configure Jira with a MySQL database using Docker containers. The learner gained practical knowledge of connecting application services with databases and managing container-based environments. The experiment helped the learner understand the role of Docker, database configuration, and JDBC drivers in building a functional multi-service application.